DBMS ASSIGNMENT

ASSIGNMENT 3

Database Design Analysis

Project: Library Loan Management System

Database: MySQL 8.0

Objective

Analyze the Library Loan Management System database design, identify functional dependencies and anomalies, and recommend suitable improvements.

Note: outputs are expected results based on the included sample data. Run the SQL in your MySQL environment and insert your own screenshots if required by your instructor.

1. Project Overview

The Library Loan Management System manages library members, book editions, authors, physical copies, and loans. It supports catalogue searches, co-authored titles, copy inventory, borrowing history, due dates, returns, and overdue-loan reporting.

2. Proposed Relational Design

Members(MemberID, FullName, Email, Phone, JoinedOn, MembershipStatus)
Books(BookID, ISBN, Title, Edition, PublicationYear, Category)
Authors(AuthorID, AuthorName)
BookAuthors(BookID, AuthorID)
BookCopies(CopyID, BookID, AccessionNumber, ShelfLocation, ConditionStatus)
Loans(LoanID, MemberID, CopyID, BorrowedOn, DueOn, ReturnedOn, OpenCopyID)

Primary keys identify rows. BookAuthors has a composite primary key (BookID, AuthorID). Foreign keys connect Books to BookCopies, Books and Authors through BookAuthors, and Members and BookCopies to Loans. A book may have multiple authors and each author may write multiple books; BookAuthors resolves this many-to-many relationship.

3. Functional Dependencies

MemberID → FullName, Email, Phone, JoinedOn, MembershipStatus
BookID → ISBN, Title, Edition, PublicationYear, Category
AuthorID → AuthorName
(BookID, AuthorID) identifies one author-to-book link.
CopyID → BookID, AccessionNumber, ShelfLocation, ConditionStatus
LoanID → MemberID, CopyID, BorrowedOn, DueOn, ReturnedOn

AccessionNumber is unique for each physical copy. OpenCopyID is generated as CopyID when a loan is open and NULL after return. Its UNIQUE constraint prevents the same copy from having two simultaneous open loans. These dependencies reflect the stated library rules.

4. Redundancy and Anomalies

A poorly designed single table might repeat member details for every loan, repeat book titles for every physical copy, and repeat author names for each book loan.

Update anomaly: changing a member's email in multiple loan rows can leave inconsistent values.
Insertion anomaly: a new book or author could not be stored until a loan exists.
Deletion anomaly: deleting the last loan for a book could accidentally remove the only stored title or author details.
Redundancy: book, member, and author information would be repeated unnecessarily.

5. Normalization Analysis

First Normal Form (1NF): every column stores one value. Authors are not stored as comma-separated text in Books; each author relationship is a row in BookAuthors.
Second Normal Form (2NF): BookAuthors contains only the composite key and has no non-key attribute dependent on only part of that key. Other entity tables use single-column primary keys.
Third Normal Form (3NF): member details depend on MemberID, title details on BookID, author names on AuthorID, and copy details on CopyID. These facts are separated to avoid transitive dependencies. The design is intended to satisfy 3NF under the listed business rules.

6. Improvements and Controls

1. Use primary and foreign keys to protect relationships.
2. Enforce unique member emails and accession numbers.
3. Keep authors in Authors and connect them using BookAuthors.
4. Use CHECK constraints for membership status, copy condition, and loan dates.
5. Use OpenCopyID with a unique constraint to stop duplicate simultaneous loans of one copy.
6. Index foreign-key columns and common search fields if performance requires it.
7. Define loan duration, renewal, fine, reservation, and lost-book policies before extending the schema.
8. Restrict access to member contact information and maintain backups.

7. Conclusion

The six-table design stores each type of fact in its appropriate place, reduces redundancy, and protects relationships. It is a normalized foundation for the SQL implementation in Assignments 5–9.